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RESULTS

- Survival models incorporating censored covariates are useful for analyzing disease progression in settings where observation can end before initial presentation of symptoms.
- Modeling a censored response and a censored covariate is challenging in this context, when ordering constraints lead to a dependent censoring mechanism.
- We develop SPARTACCUS (SemiPArametric Right-censored event Time and Censored Covariate as USual), a Bayesian joint modeling approach incorporating semiparametric Cox proportional hazards models for both the right-censored response and censored covariate.
- We implement a computationally efficient MCMC sampler that avoids data augmentation and can handle high-censoring rates.
- We compare SPARTACCUS to complete case analysis in high-censoring settings across a range of sample sizes and hazard shapes.
- We apply SPARTACCUS to the PREDICT-HD clinical trial with HD-ISS Huntington's disease staging.



Figure 1: RMSE percentage improvement of SPARTACUS over complete-case CoxPH for Stage I-to-Stage II duration model parameters (α , γ). We conduct simulation studies where both censored response and censored covariate are generated from a Weibull proportional hazards model under a 3^2 factorial design, with factors: (i) sample size $n \in \{100, 1000, 2500\}$; (ii) covariate censoring rate $r \in \{0.20, 0.40, 0.80\}$; (iii) Weibull shape $k \in \{0.75, 1.0, 1.5\}$. We use 100,000 MCMC iterations with 50,000 burn in iterations. Cells are coloured by RMSE improvement, with positive values (teal) indicating SPARTACUS achieves lower RMSE. Numeric labels show RMSE for SPARTACUS (bold) and complete-case.

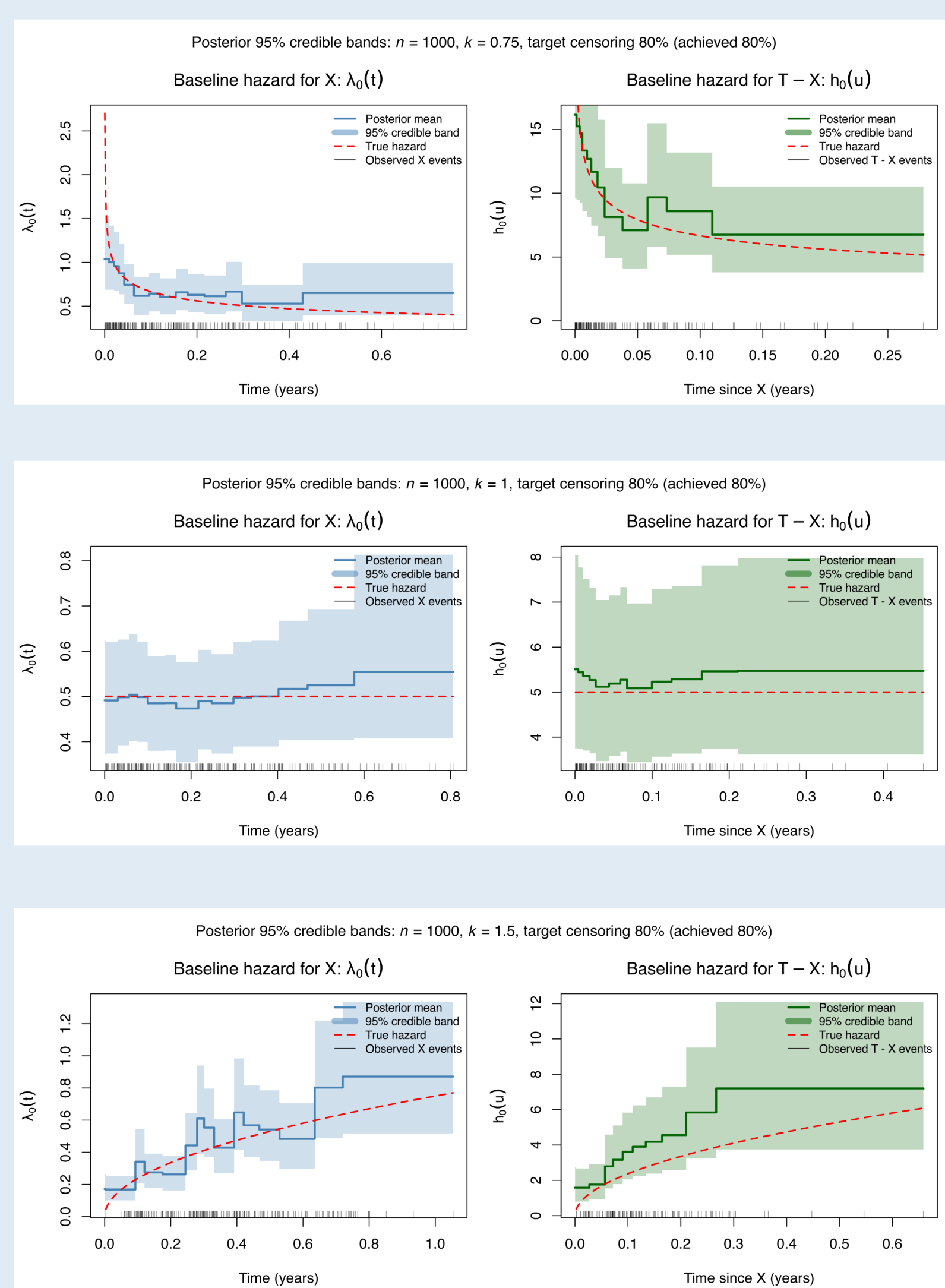


Figure 2: Posterior mean (solid), 95% credible band (shaded), and Weibull truth (red dashed) for the baseline hazards $\lambda(t)$ (left) and $h(t)$ (right) for 3 simulated datasets with $n = 1000$, target covariate censoring rates 80%, $k = \{0.75, 1, 1.5\}$ (top to bottom).

Semiparametric Joint Survival Model for a Right Censored Covariate

We denote by T_i , X_i and C_i an event time response, an event time covariate, and censoring variable, respectively, for subject $i = 1, \dots, n$. Both T_i and X_i can be right-censored by C_i and are ordered $X_i \leq T_i$, so that either (i) they are both right-censored, (ii) X_i is completely observed and T_i is right-censored, or (iii) both are completely observed. Accordingly, we define notation for censored data

$$\begin{aligned} T_i^* &= \min(T_i, C_i), & X_i^* &= \min(X_i, C_i), \\ \delta_{X,i} &= I\{X_i \leq C_i\}, & \delta_{T,i} &= I\{T_i \leq C_i\}. \end{aligned}$$

In our motivating study, X_i represents the time to first detectable subcortical neurodegeneration (HD-ISS Stage I) and T_i represents the time to first clinical sign or symptom (HD-ISS Stage II) for subjects with > 40 CAG repeats enrolled in the PREDICT-HD study.

We further denote by $\mathbf{Z}_i \in \mathbb{R}^p$ a vector of uncensored covariates representing baseline regional cortical volumes, and our interest lies in regressing T_i on both X_i and $\mathbf{Z}_i$.

We assume that $(T_i, X_i) \perp C_i \mid \mathbf{Z}_i$, let $U_i = T_i - X_i$, and assume a joint Cox proportional hazard models for X_i and U_i .

$$\begin{aligned} h_X(x \mid \mathbf{Z}) &= \lambda_0(x) \exp(\mathbf{Z}^\top \boldsymbol{\beta}), \\ p(u \mid X, \mathbf{Z}) &= h_0(u) \exp(X\gamma + \mathbf{Z}^\top \boldsymbol{\alpha}). \end{aligned}$$

The joint survival censored covariate regression model can be viewed as a three-state model with a progressive state-structure, Stage-0 $\rightarrow$ Stage-I $\rightarrow$ Stage-II. In this context Markovian transitions between states correspond to the special case where $\gamma = 0$.

Flexible piecewise constant forms for the baseline hazard functions

$$\begin{aligned}\lambda_0(x) &= \sum_{j=1}^{J_\lambda} \lambda_j I\{x \in (s_{j-1}^\lambda, s_j^\lambda]\}, \\ h_0(u) &= \sum_{j=1}^{J_h} h_j I\{u \in (s_{j-1}^h, s_j^h]\}.\end{aligned}$$

Letting $\psi_j^\lambda = \log \lambda_j$ and $\psi_j^h = \log h_j$, we assign first-order random-walk smoothing priors

$$\begin{aligned}\psi_j^\lambda &= \psi_{j-1}^\lambda + a_j^\lambda, & a_j^\lambda &\stackrel{iid}{\sim} N(0, \sigma_\lambda^2), & \psi_1^\lambda &\sim N(0, A_\psi^2), \\ \psi_j^h &= \psi_{j-1}^h + a_j^h, & a_j^h &\stackrel{iid}{\sim} N(0, \sigma_h^2), & \psi_1^h &\sim N(0, A_\psi^2).\end{aligned}$$

The regression coefficients are assigned normal priors,

$$\alpha_{z_j} \stackrel{iid}{\sim} N(0, A_\alpha^2), \quad \beta_{z_j} \stackrel{iid}{\sim} N(0, A_\beta^2), \quad \gamma \sim N(0, A_\gamma^2).$$

and the smoothing parameters for the baseline hazards are assigned half-Cauchy priors,

$$\sigma_\lambda, \sigma_h \overset{ind}{\sim} \text{Half-Cauchy}(0, A_\sigma).$$

with $A_\psi, A_\gamma, A_\beta, A_\alpha, A_\sigma$ being fixed scale hyperparameters chosen to yield weakly informative prior specifications. In all analyses we set $A_\psi = A_\alpha = A_\beta = A_\gamma = 2$ and $A_\sigma = 25$.

Likelihood and Posterior Distribution

The observed-data likelihood then takes the form

$$\begin{aligned} L(\theta) = & \prod_{i:\delta_{X,i}=1, \delta_{T,i}=1} f_{T,X|Z}(t_i^*, x_i^* \mid \mathbf{z}_i, \theta) \\ & \times \prod_{i:\delta_{X,i}=1, \delta_{T,i}=0} \int_{t_i^*}^{\infty} f_{T,X|Z}(t, x_i^* \mid \mathbf{z}_i, \theta) dt \\ & \times \prod_{i:\delta_{X,i}=0, \delta_{T,i}=0} \int_{C_i}^{\infty} f_{T,X|Z}(t, x \mid \mathbf{z}_i, \theta) dt dx. \end{aligned}$$

The cumulative hazard terms are evaluated directly under the piecewise-constant baseline hazards and thus the observed data likelihood has a closed form expression. As all three terms of the observed data likelihood have closed forms, data augmentation is not required for censored observations.

The log-baseline hazard vectors $\{\psi_{\lambda j}^1\}_{j=1}^{J_{\lambda}}$ and $\{\psi_{\theta j}^1\}_{j=1}^{J_{\theta}}$ are updated in blocks using automated factor slice sampling under the random-walk prior along with the regression coefficients in their respective hazard models. The smoothing standard deviations σ_{λ} and σ_{θ} are then updated using univariate slice sampling. The posterior sampling algorithm is implemented in R-NIMBLE (R 4.4.0 - NIMBLE 1.3.0).

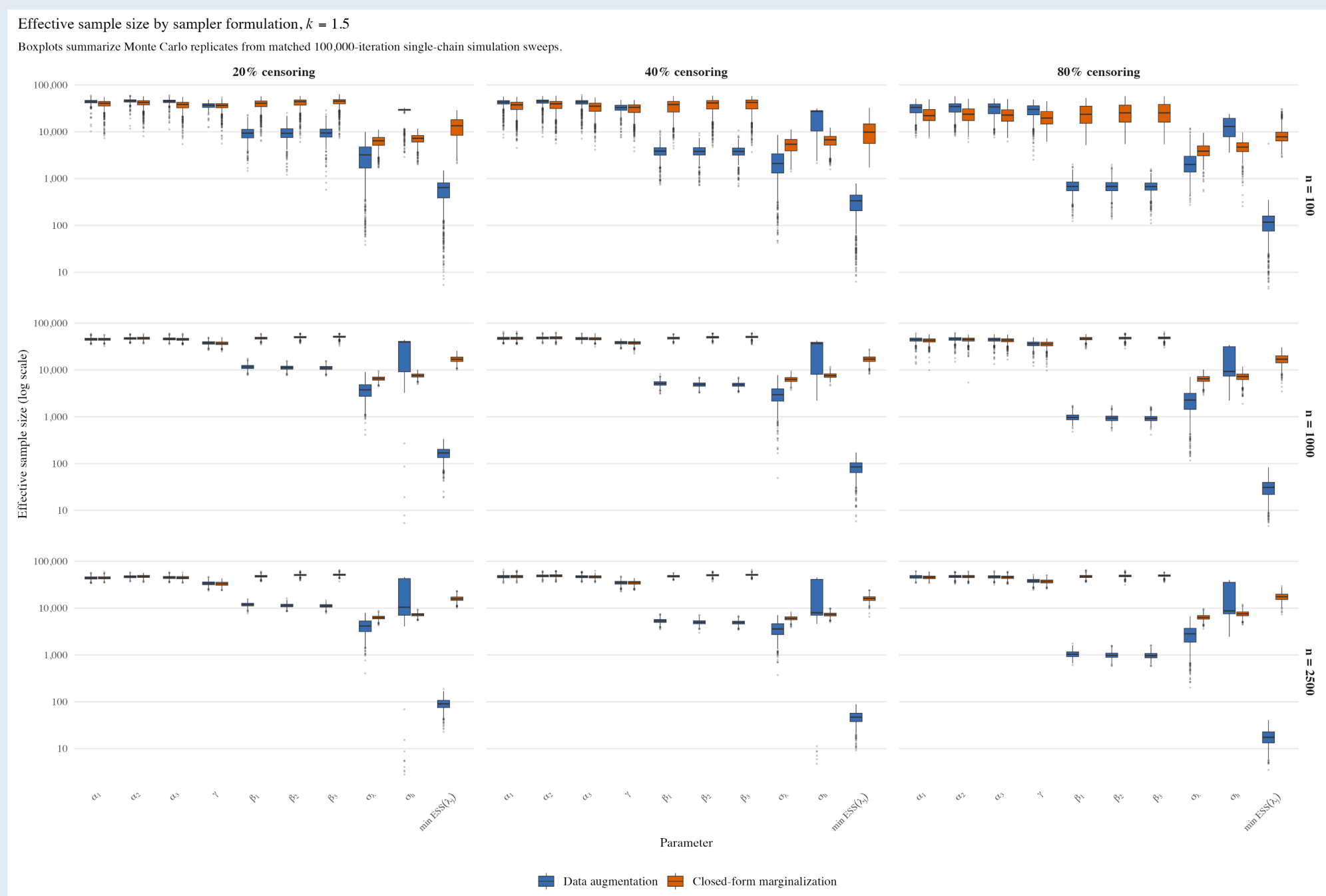


Figure 3: Effective sample size distributions across simulation replicates for $k = 0.75$ under the adaptive square-root breakpoint rule. Panels correspond to target censoring rates; boxplots summarize one-chain ESS across replicates for each parameter and sample size. The min ESS(λ_i) and min ESS(h_i) entries are per-replicate minima over the corresponding baseline-hazard pieces.

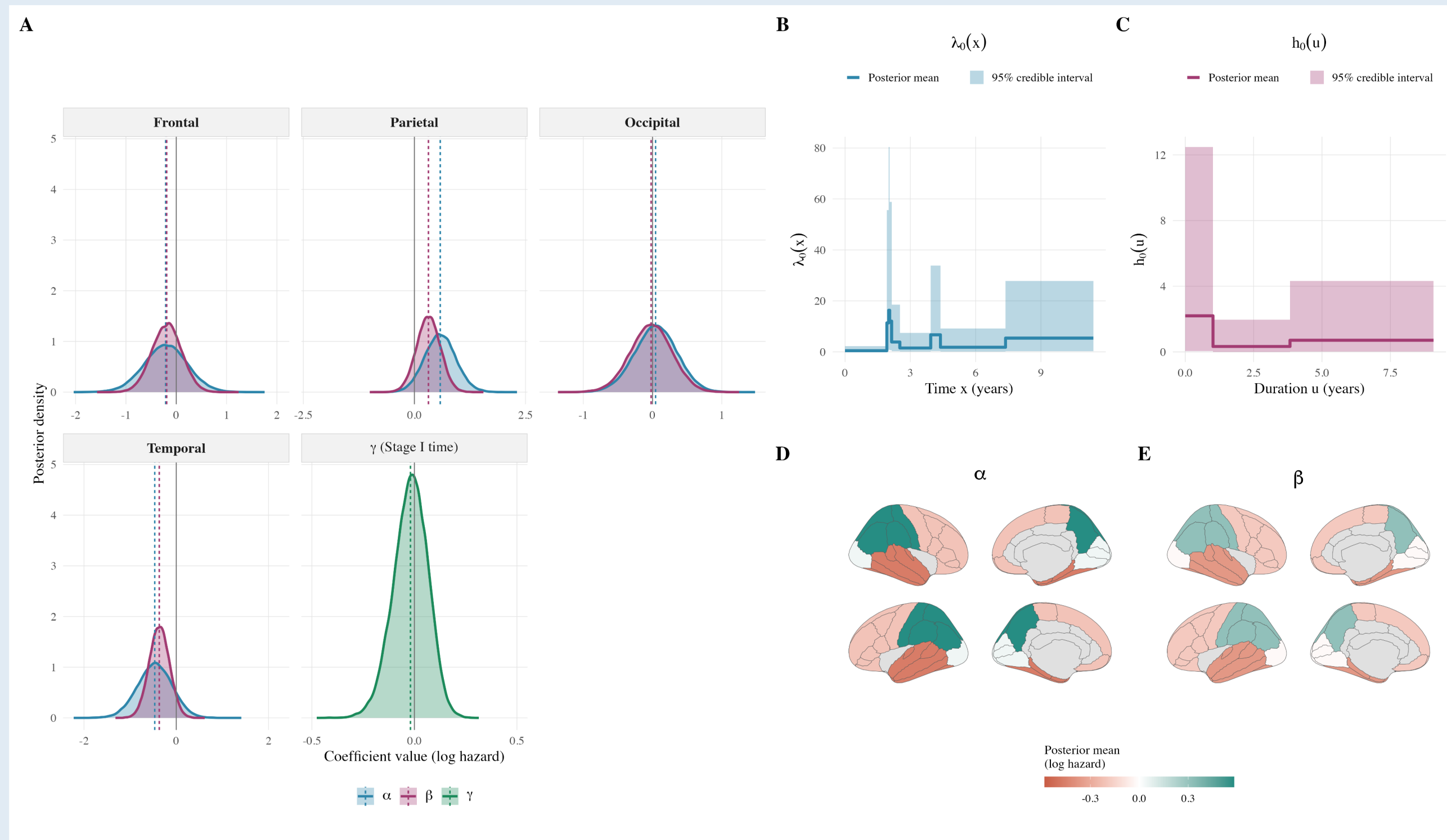


Figure 4: SPARTACUS application to PREDICT-HD (n = 149). (A) Posterior densities for α (Stage I to Stage II duration, blue), β (time to Stage I, pink), and γ (Stage I adjustment, green). Dashed vertical lines mark the posterior means. (B) Estimated baseline hazard $\lambda_0(t)$ for time to Stage I (posterior mean and 95% credible band), with elevated early follow-up hazards and a later increase. (C) Estimated baseline hazard $\lambda_0(u)$ for Stage I-to-Stage II duration, front-loaded after Stage I. (D)–(E) Brain surface maps of posterior means for α and β , respectively.

Figure 1 compares RMSE for SPARTACCUS and the complete-case estimator of (α, γ) . SPARTACCUS has lower RMSE in all 12 settings with $n = 100$ and 80% covariate censoring. Complete-case RMSE reaches 13.6 for α and 101.2 for γ , compared with maxima of 0.488 and 1.152 for SPARTACCUS. This regime resembles the Huntington's disease application ($n = 149$; 56% double censoring). Figure 2 shows three representative high-censoring simulations across baseline hazard shapes. The true hazards remain within the 95% credible intervals, indicating reasonable hazard recovery. Figure 3 assesses MCMC mixing across replicates. Most parameters have effective sample sizes above 10,000; mixing is weakest for the random-walk innovation parameters and some piecewise-exponential hazard components. Figure 4 summarizes the application results. The Stage-I baseline hazard, $\lambda_0(t)$, is elevated early and rises late in follow-up, while the Stage-I-to-Stage-II hazard, $h_0(u)$, is highest during the first year. Parietal volume is positively associated with Stage-II progression ($\hat{\alpha}_{\text{parietal}} = 0.581$; HR = 1.79), whereas temporal volume is negatively associated with Stage-I onset ($\hat{\beta}_{\text{temporal}} = -0.368$; HR = 0.69) and later progression ($\hat{\alpha}_{\text{temporal}} = -0.467$; HR = 0.63). The Stage-I timing effect is near zero ($\hat{\gamma} = -0.019$), suggesting approximate conditional hazard independence.

DISCUSSION AND CONCLUSION

- Our simulation studies demonstrate that the proposed joint model improves estimation of the Stage II model parameters in small-sample, high-censoring settings compared with complete-case Cox regression, while performing similarly in settings where complete-case analysis is already efficient.
- In the PREDICT-HD analysis, SPARTACCUS identifies posterior shifts in the parietal and temporal lobes that can be used to identify brain stimulation targets and our estimate of γ suggests Markovian transitions between disease states.
- Our basic framework in this study can be extended to more fine scale neuroimaging data incorporating a spatial prior such as a conditional autoregressive prior for the regression coefficients of the regional neuroimaging measures in a space-varying regression Gamerman et al. (2003).
- The spatial extension and performance in data-sparse settings suggests broader application to brain–disease association studies. Future work will consider interval-censoring and extensions that relax the proportional hazards assumption for censored covariates.

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REFS

